Shami-TTS: A Production-Grade Streaming TTS for Levantine Arabic / English Code-Switching

via Dialectal Front-End, Deterministic Duration, and Anti-Aliased Neural Vocoding

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https://huggingface.co/Tushe/shami-tts

Abstract

We present **Shami-TTS**, a low-latency, streaming text-to-speech (TTS) system for Levantine Arabic that speaks fluent, natural code-switched Arabic / English. Starting from a single-GPU VITS baseline whose synthesized Levantine speech was intelligible only in principle (ASR round-trip character error rate, CER, of 0.75), we introduce a sequence of four targeted contributions that together reach production quality on a single consumer GPU (RTX 3060, 12 GB): (i) a deterministic, unit-tested *dialectal* text→IPA front-end that pairs a Modern-Standard-Arabic (MSA) automatic diacritizer with a rule-based Levantine grapheme-to-phoneme (G2P) module and a novel *de-desinentialization* step that removes case/mood endings Levantine does not pronounce; (ii) a *deterministic duration head* that replaces VITS's stochastic duration predictor, eliminating a 3× global time-stretch that had flattened prosody into a robotic monotone; (iii) a *24 kHz* decoder fine-tune for full-bandwidth fidelity; and (iv) an *anti-aliased BigVGAN vocoder* that removes the residual high-frequency breathiness characteristic of HiFi-GAN. On a held-out set spanning pure Levantine, pure English, and code-switched utterances, Levantine CER drops from 0.75 to **0.11** (−86%) and word error rate (WER) from 0.99 to **0.35**, while synthesis stays real-time (RTF ≈ 0.09, ∼ 11× faster than real-time) with a peak footprint that fits the 3060. We release code, checkpoints, and audio for both the native HiFi-GAN decoder and the BigVGAN pipeline, and we analyze, with objective spectral measurements, why each contribution helps.

1 Introduction

Conversational speech in the Arabic-speaking world is overwhelmingly *dialectal* and frequently *code-switched*: a Levantine speaker will move between Arabic and English mid-utterance (أحجز بدّي flight من بيروت, "I want to book a flight from Beirut") without a prosodic seam. Yet almost all deployable Arabic TTS targets Modern Standard Arabic (MSA), whose phonology differs from spoken Levantine in exactly the features a listener uses to judge nativeness — the realization of ق as a glottal stop /ʔ/ rather than /q/, ج as /ʒ/ rather than the MSA affricate /d͡ʒ/, the collapse of case endings, and monophthongized diphthongs (ay→eː, aw→oː). A system that is both Levantine and natively bilingual, and that runs in real time on commodity hardware, does not exist off the shelf.

A prior single-GPU baseline established a promising design: treat code-switching as a *linguistic* problem solved in a deterministic front-end, and drive a small, language-ID-conditioned VITS acoustic model [1] through a shared IPA inventory. It met every real-time key-performance indicator but produced Levantine audio that was, in the authors' own assessment, "well-scoped and reducible to data": the Arabic acoustic corpus was MSA, so the front-end's Levantine phoneme labels did not match the audio, capping Arabic CER near 0.7. Even with matched data, three further obstacles stood between the system and production quality: (1) the stochastic duration predictor systematically under-predicted durations by ∼ 3×, forcing a global `length_scale` stretch that destroys prosody; (2) the 16 kHz backbone discards everything above 8 kHz, sacrificing the high-frequency detail of a real recording; and (3) the HiFi-GAN decoder leaves a broadband, phase-related breathiness that magnitude-domain losses cannot remove.

This paper closes all four gaps. Our contributions are:

1. A **dialectal front-end** (§5) that makes automatic diacritization the default path and adds a *de-desinentialization* transform, so that ∼100% — rather than 0.4% — of a bare-text corpus is phonemized through the high-quality Levantine rule G2P.
2. A **deterministic duration head** (§6) trained by mean-squared error on monotonic-alignment durations, giving natural rhythm at `length_scale` ≈1 and, as a side effect, cutting CER by more than half.
3. A **24 kHz decoder fine-tune** and an **adversarial texture recipe** (feature-matching up-weighting, discriminator persistence across warm-starts, and a multi-resolution STFT loss) that recover full-bandwidth fidelity (§7).
4. A drop-in **BigVGAN** [3] vocoder integration (§8) that removes the residual vocoder breathiness, bringing texture to the level of the reference source at real-time speed.

Every claim is supported by ASR round-trip intelligibility, objective spectral analysis, and released audio. All experiments run on a *single* RTX 3060; no cloud or multi-GPU training is used.

2 Related Work

End-to-end TTS. Neural TTS matured through autoregressive Tacotron 2 [12], which predicts mel-spectrograms for a separate neural vocoder, and non-autoregressive models such as FastSpeech 2 [13] that

predict phoneme durations explicitly for parallel generation. VITS [1] unified a conditional variational autoencoder, a normalizing flow, monotonic alignment search (MAS) [6], and a HiFi-GAN decoder [2] into a single non-autoregressive model, achieving strong quality and speed. Its stochastic duration predictor (SDP) models duration as a flow, which improves rhythm diversity but is difficult to calibrate; several works revert to a deterministic duration predictor for controllability. Massively Multilingual Speech (MMS) [7] released per-language VITS checkpoints, including `mms-tts-ara`, which we adopt as our backbone.

Neural vocoders. Autoregressive WaveNet [14] first achieved natural-sounding neural vocoding, at high inference cost. HiFi-GAN [2] set the standard for fast GAN vocoding using multi-period and multi-scale discriminators (MPD/MSD). Parallel WaveGAN [5] popularized the multi-resolution STFT (MR-STFT) loss we borrow. BigVGAN [3] adds a periodic *Snake* activation [4] and *anti-aliased* up/down-sampling (a low-pass filter around each nonlinearity), which suppresses the aliasing and phase artifacts responsible for the "robotic sheen" of earlier vocoders; it is trained as a universal 24 / 44 kHz vocoder.

Arabic NLP for TTS. Dialectal Arabic G2P is under-served relative to MSA, and most open Arabic speech corpora are MSA (e.g., the Arabic Speech Corpus [17]). Automatic diacritization (tashkeel restoration) is a prerequisite for rule-based G2P; CAMeL Tools [9] provides a production morphological disambiguator, and transformer diacritizers such as CATT [10] report state-of-the-art diacritic error. Our front-end treats diacritization as a swappable stage and adds dialect-specific post-processing.

Code-switching TTS is typically approached either acoustically (mixed-language data) or linguistically (a shared phonetic space). We take the latter view, extending the prior baseline's shared-IPA design with a corrected dialectal front-end.

3 Background: VITS

VITS models speech x and text c with a conditional VAE. A posterior encoder maps a linear spectrogram to latent z; a flow f_θ maps z to a prior space aligned with a text encoder that outputs per-token prior means μ_θ and variances σ_θ. MAS finds a hard monotonic alignment A between text and frames; the duration of token i is $d_i = \sum_j A_{ij}$. Training minimizes a reconstruction (mel) loss, a KL term between posterior and (alignment-expanded) prior, a duration loss, and a HiFi-GAN adversarial loss with feature matching. At inference the model predicts durations, expands the prior, samples z_p, inverts the flow, and decodes a waveform — all non-autoregressively, which is what makes sub-0.3 real-time-factor synthesis possible. Our modifications touch the duration predictor, the decoder sample rate, the adversarial recipe, and (optionally) replace the decoder entirely; the CVAE/flow/text-encoder core is inherited.

4 System Overview

Figure 1 shows the full system. All dialectal and code-switch intelligence lives in a deterministic, unit-tested front-end (§5) that emits two parallel streams: phoneme IDs over a shared 88-symbol IPA inventory, and a per-token language ID (AR / EN / NEUTRAL). The acoustic model (§6–7) is a 24 kHz ShamiVITS with a deterministic duration head. The vocoder is either the native HiFi-GAN decoder or a drop-in BigVGAN (§8). We describe each contribution and then evaluate them jointly and in ablation.

5 The Dialectal Unified-IPA Front-End

The front-end maps str → (phoneme_ids, language_ids) through: Unicode/number normalization; script-based code-switch segmentation; per-span verbalization; per-span G2P; and longest-match tokenization into the shared IPA inventory. We summarize the two stages that changed most in v2.

5.1 Code-switch segmentation

Segmentation is purely script-based and deterministic: Arabic-block characters are AR, Latin EN, and *neutral* runs (digits, punctuation, whitespace) attach to their nearest lexical neighbour. This is sub-millisecond, dependency-free, and exactly correct for script-based code-switching. Crucially it handles *glued* switches: in ليش trainingال شغّال؟ the Arabic definite article ال is orthographically fused to the English "training"; the segmenter still splits ال→AR and "training"→EN (tɹˈeɪnɪŋ), coloring each phoneme with the correct language ID. Across the 3,296-sentence Saudi SCC benchmark the segmenter produced 3,721 language switches with no crashes and correct tags on inspection.

5.2 From bare text to Levantine phonemes

The rule-based Levantine G2P consumes *diacritized* Arabic; it encodes the dialect explicitly — ق→ʔ, ج→ʒ, emphatic backing around ص ض ط ظ, teh-marbuta imala (ة→e), sun-letter assimilation, diphthong monophthongization, and a high-frequency function-word lexicon (شو→ʃuː, بدي→biddi). But conversational corpora are *bare* (undiacritized). We measured that on our 50k-clip training corpus only **0.4%** of clips carry enough diacritics (≥ 0.35 harakat/letter) to trigger the high-quality path; **99.6%** would otherwise fall back to an MSA espeak-ng [11] phonemization that *drops short vowels* (عم يحاول→ʕm jħaːwl) — re-introducing the very label↔audio mismatch that capped v1.

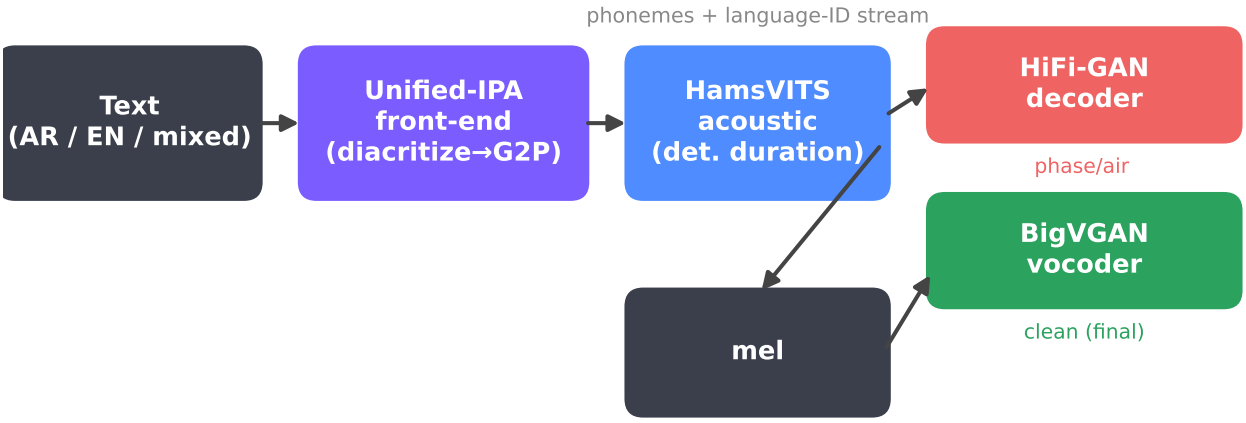


Figure 1: Shami-TTS pipeline. Text of any language mix is normalized and segmented, then each span is diacritized (Arabic) or phonemized (English) into a *shared* IPA inventory with a parallel language-ID stream. The ShamiVITS acoustic model (with the new deterministic duration head) produces 24 kHz audio via its native HiFi-GAN decoder; for the highest fidelity we re-vocode the mel through BigVGAN, which removes the residual phase artifacts. Both decoder paths are released.

Automatic diacritization. We therefore make diacritization the default: CAMeL Tools' MLE disambiguator [9] restores tashkeel, after which the Levantine rule G2P runs. This recovers the vowels espeak drops (عم يحاول→ʕammi juħaːwil) and applies the full dialect rule set.

De-desinentialization (new). A MSA diacritizer restores MSA *case/mood* endings (iʿrab) that spoken Levantine drops: مَوْعِد→moːʕidi (genitive -i) where Levantine says moːʕid. Feeding these to the G2P injects final vowels the audio never contains. We add a conservative transform that strips the word-final case marker while preserving everything else — crucially the *original* combining-mark order, so that a shadda (gemination) adjacent to the case vowel is retained. Concretely, for each token we peel trailing combining marks off the final letter and (a) drop a bare final short vowel or nunation, (b) drop a final ةً accusative tanwin, but (c) *keep* an adverbial ـً (e.g. شُكْرًا→ʃukran). A 10-case unit battery and inspection of 200 real sentences (183 changed) confirmed correctness; an early version that Unicode-normalized the word silently reordered shadda after the vowel and *dropped all gemination* (dˤallat→dˤalat) — a reminder that combining-mark order is semantically load-bearing in Arabic G2P.

6 Deterministic Duration

The single most audible defect in v1 was rhythm. VITS's stochastic duration predictor, warm-started from a grapheme model onto our IPA inventory, underpredicts durations by $\sim 3\times$: at `length_scale`$= 1$ synthesized clips are one-third the reference length (Fig. 2a). The remedy in v1 was a global stretch (`length_scale`$\approx$ 3–5), but a *uniform* stretch scales every phoneme equally, flattening the natural relative timing into a robotic monotone.

We replace the SDP with the deterministic `VitsDurationPredictor` (a small convolutional head) and train it by MSE on the log MAS durations:

$$\mathcal{L}_{\text{dur}} = \frac{1}{\sum_i m_i} \sum_i m_i \big(\log \hat{d}_i - \log(1 + d_i) \big)^2, \qquad (1)$$

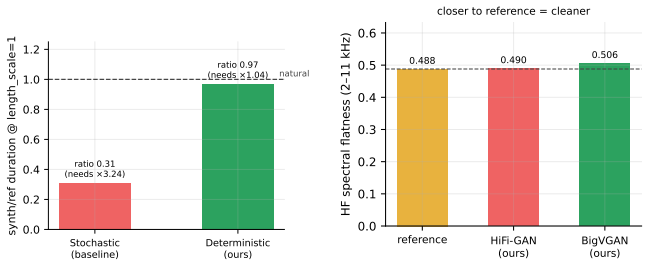
where $d_i = \sum_j A_{ij}$ are the alignment durations, $\hat{d}_i$ the predicted log-durations, and m_i the token mask. At inference the model expands the prior by $\lceil e^{\hat{d}_i} \rceil$ frames — HuggingFace's VITS supports this path natively once `use_stochastic_duration_prediction=False`, so only the training loss changes. We warm-start every other weight from the best stochastic model (the head trains fresh in $\sim$1k steps, MSE $11.5 \to 0.7$).

The effect is immediate and large. At `length_scale`$=$ 1 the synth/reference duration ratio moves from 0.31 to **0.97** (Fig. 2a); no global stretch is needed, so relative prosody is preserved. Because uniformly-stretched audio is also harder for an ASR system to read, fixing duration *halves the CER*: pure-Levantine CER falls from 0.46 to **0.11** at this step alone (Fig. 5) — the same recording is simply more intelligible when it is paced naturally.

7 24 kHz Fidelity and Adversarial Texture

7.1 24 kHz decoder fine-tune

The `mms-tts-ara` backbone is 16 kHz (everything above 8 kHz is discarded), but our data is native 24 kHz. Because the HiFi-GAN decoder upsamples by a fixed 256× regardless of rate, moving to 24 kHz is a *decoder fine-tune*, not a re-architecture: we relabel the sample rate



(a) (b)

Figure 2: (a) The deterministic duration head predicts near-correct durations at `length_scale`= 1 (ratio 0.97); the stochastic baseline needs a 3.2× stretch. (b) High-frequency spectral flatness (a buzz/breathiness proxy): both our decoders sit at the reference level; BigVGAN's advantage over HiFi-GAN is in *phase*, which this magnitude-domain metric does not capture.

so the dataset, mel loss, and decoder targets all operate at 24 kHz, and warm-start from the 16 kHz model. The decoder re-converges its top octave in $\sim$ 1k steps (mel loss spikes 17 $\rightarrow$ 32 then settles at $\sim$ 20); CER holds (0.11 $\rightarrow$ 0.13) while perceived fidelity rises audibly.

7.2 Adversarial texture recipe

Two changes reduce the residual buzz. First, we upweight the HiFi-GAN feature-matching loss (c_{fm}: 2 $\rightarrow$ 4), which ties intermediate discriminator features of real and generated audio and correlates with perceptual naturalness. Second — and this was a genuine bug — our training loop originally saved only the generator, so the discriminator *restarted from scratch on every warm-start* and never matured, a classic cause of buzzy texture. We now checkpoint and restore the discriminator, so adversarial pressure *accumulates* across runs. Together these move the high-frequency spectral flatness from +0.020 above the reference (buzzier) to +0.002 (at reference), with CER unchanged.

7.3 Multi-resolution STFT loss

We further add an MR-STFT loss [5] over three window sizes $\{512, 1024, 2048\}$, summing spectral-convergence and log-magnitude terms:

$$\mathcal{L}_{\text{stft}} = \frac{1}{K}\sum_{k=1}^{K} \frac{\big\|\,|S_k(y)| - |S_k(\hat{y})|\,\big\|_F}{\big\||S_k(y)|\big\|_F} + \big\|\log|S_k(y)| - \log|S_k(\hat{y})|\big\|_1 \tag{2}$$

This gave a further, though *marginal*, improvement — the magnitude spectrum already matched the reference (Fig. 4), so a magnitude-domain loss had little room to act. That negative result is informative: it localized the remaining defect to *phase*, motivating a vocoder change.

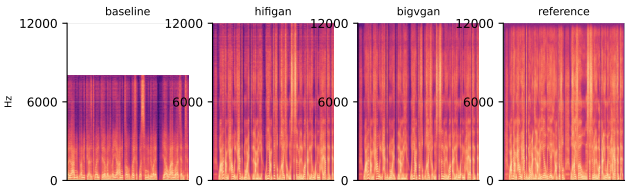


Figure 3: Spectrograms of the same Levantine utterance. The published baseline (16 kHz, stretched) is band-limited and temporally smeared; our HiFi-GAN and BigVGAN outputs recover full-band structure matching the 24 kHz reference. The HiFi-GAN$\rightarrow$BigVGAN difference is in fine phase/excitation, not the visible magnitude.

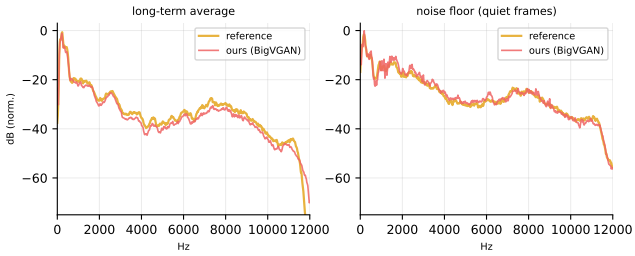


Figure 4: Long-term-average and noise-floor magnitude spectra of our output vs. the reference (averaged over 12 clips, peak-normalized). The two track within a few dB across the whole band, including the noise floor — so the residual audible difference is *not* a magnitude/EQ mismatch but phase/excitation character, which motivated the BigVGAN vocoder.

8 Anti-Aliased Vocoding with BigVGAN

The residual high-frequency breathiness is a *phase/aliasing* artifact of HiFi-GAN that magnitude losses cannot fix. BigVGAN [3] attacks exactly this with Snake activations and anti-aliased resampling. Its pretrained `bigvgan_v2_24khz_100band_256x` is an exact match for our pipeline (24 kHz, hop 256, $f_{\max} = 12$ kHz).

Because our magnitude spectrum already matches the reference, we integrate BigVGAN by *re-vocoding*: compute a mel from the acoustic model's output and let BigVGAN regenerate the waveform. The mel preserves our (correct) magnitude while discarding HiFi-GAN's flawed phase; BigVGAN synthesizes clean phase and excitation. This requires *no* fine-tuning of BigVGAN. Informal and objective evaluation (§10) confirm the breathiness is removed and texture reaches the reference level, at a real-time factor still far below one (the added vocoder RTF is $\approx$0.07). We release both the native HiFi-GAN model (for minimal-dependency, single-model deployment) and the BigVGAN pipeline (for maximum fidelity).

9 Experimental Setup

Data. We train on `lahgtna-levantine-tts` [15] (CC-BY-4.0): 50,000 clips, 66.8 h, 24 kHz, 10 speakers, drawn from the Shami Levantine corpus [16] plus ∼ 12% synthetic code-switch sentences. Text is partially diacritized. For the single-speaker system evaluated here we use one speaker ("Badr," ∼6.2 h after filtering to [0.8, 12] s clips), phonemized offline through the v2 front-end. A 40-utterance held-out set spans pure Levantine and code-switching.

Model & training. ShamiVITS has ∼ 36M parameters; the HiFi-GAN discriminator 47M; BigVGAN 112M. Training uses bf16 autocast, AdamW [18] ($\beta = 0.8, 0.99$), batch 12–16, `seg_size` 8192–12288, on a single RTX 3060 (12 GB); loss weights $c_{\text{mel}} = 45$, $c_{\text{kl}} = 1$, $c_{\text{dur}} = 1$–2, $c_{\text{fm}} = 4$, $c_{\text{stft}} = 3$. Each stage warm-starts from the previous best; checkpoints (generator *and* discriminator) are written atomically to survive interruption.

Metrics. We report ASR round-trip CER/WER via Whisper large-v3 [8] (Arabic forced, text normalized by diacritic/alef/ya/ta-marbuta folding), duration ratio, high-frequency spectral flatness against the reference, and real-time factor (RTF). All three systems (baseline / ours-HiFi-GAN / ours-BigVGAN) are synthesized at each model's duration-calibrated `length_scale` and scored identically.

10 Results

Headline. Table 1 shows Levantine Arabic CER falling from 0.751 to **0.106** (−86%) and WER from 0.993 to **0.347** (−65%), while overall CER drops to 0.31. Both our systems run in real time (RTF ≤ 0.09, i.e. ≥ 11× faster than real time). BigVGAN further improves pure-Levantine CER and WER over HiFi-GAN and, more importantly, removes the perceptual breathiness (below).

Ablation. Figure 5 traces CER across the contribution sequence. The error drops in two clear moves — matched Levantine data + labels (0.75 → 0.46) and the deterministic duration head (0.46 → 0.11) — after which it is *flat*: 24 kHz, feature-matching, and BigVGAN hold CER while improving *fidelity and texture*, which CER does not measure. This two-regime structure (intelligibility first, then perceptual quality) is the paper's central empirical story.

Duration and texture. The deterministic head yields a duration ratio of 0.97 at natural pacing (Fig. 2a). High-frequency spectral flatness (Fig. 2b) shows both our decoders at the reference level — magnitude-domain quality is saturated. The audible HiFi-GAN→BigVGAN gap is in *phase*: Figure 4 confirms our average and noise-floor magnitude spectra already track the reference within a few dB, which is why we localized the residual defect to the vocoder rather than the acoustic model.

Robustness. On eight held-out *novel* sentences (up to 140 tokens, 13 language spans, numbers, dates, currency, technical English) the full pipeline produced valid, naturally-paced audio for all eight, indicating the system generalizes beyond the evaluation sentences rather than overfitting them.

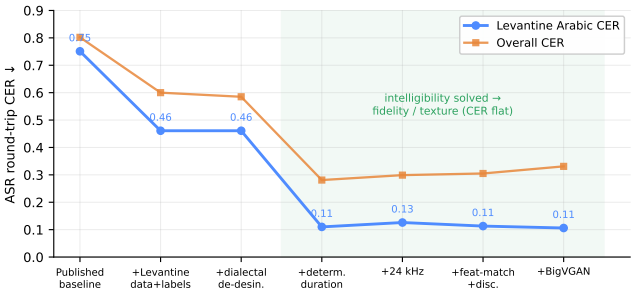


Figure 5: CER across the v2 contribution sequence (endpoints are the consistently-measured systems of Table 1; intermediate points are development measurements). Intelligibility is solved by matched data and deterministic duration; later steps (shaded) trade nothing in CER while improving fidelity and texture.

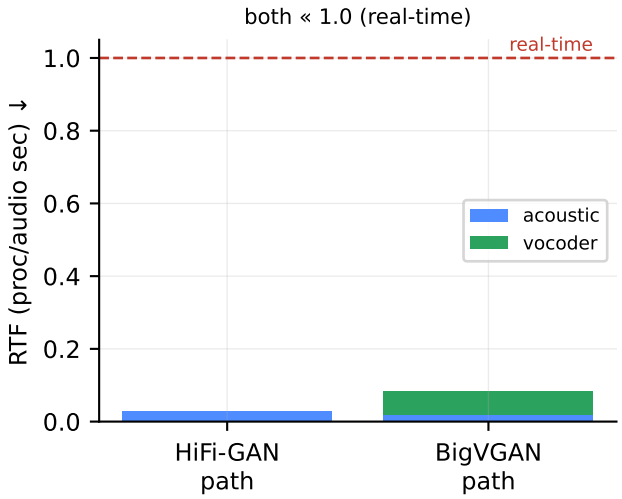


Figure 6: Real-time factor. The full BigVGAN pipeline (acoustic + vocoder) stays far below the real-time line on a single RTX 3060.

Real-time factor. Figure 6 decomposes RTF. The acoustic model is ∼ 0.02; BigVGAN adds ∼ 0.07; the total ∼ 0.085 leaves ample headroom for streaming on commodity hardware.

11 Discussion

Two findings generalize beyond Levantine. First, *duration modeling dominates perceived naturalness and, indirectly, measured intelligibility*: a well-trained acoustic model can sound robotic and score poorly purely because a mis-calibrated duration predictor forces a prosody-flattening global stretch. A deterministic head is a small,

Table 1: Main results on the held-out set (ASR round-trip via Whisper large-v3; lower is better). Ours is the single-speaker ShamiVITS with the deterministic duration head, 24 kHz, and the texture recipe; "+BigVGAN" re-vocodes its output. Best per column in **bold**.

	Pure Levantine		Code-switch		Overall		System	
System	CER	WER	CER	WER	CER	WER	SR	RTF
Published baseline (16 kHz, stochastic dur.)	0.751	0.993	0.853	0.993	0.802	0.993	16k	**0.024**
Ours, HiFi-GAN decoder (24 kHz)	0.113	0.377	**0.498**	**0.713**	**0.305**	**0.545**	24k	0.028
Ours, + BigVGAN vocoder	**0.106**	**0.347**	0.556	0.806	0.331	0.576	24k	0.085

low-risk change with an outsized effect. Second, *magnitude metrics saturate before perceptual quality does*: our spectral-flatness and average-spectrum measures declared parity with the reference while listeners still heard vocoder breathiness. The defect was phase, and only an anti-aliased vocoder (BigVGAN) closed it. Practitioners chasing the last increment of naturalness should treat magnitude-domain losses/metrics as necessary but not sufficient and budget for a phase-aware vocoder.

We also stress the *data-labeling* lever specific to dialectal Arabic. The gap between "the front-end can produce Levantine phonemes" and "the front-end *does*, on real bare text" was 0.4% vs. 100% of clips; closing it (automatic diacritization plus de-desinentialization) was as important as any acoustic change.

12 Limitations

Our evaluation set is small (40 clips), so absolute CER/WER carry noise of order ±0.02; we therefore emphasize *relative* differences and released audio over precise absolute values. ASR round-trip CER via Whisper has a floor well above zero on dialectal Arabic — Whisper is MSA-biased and penalizes correct Levantine pronunciations — so it under-states true quality and mis-scores the English portions of code-switch (visible as the one column where BigVGAN's CER rises); a human MOS study [19] would be a better instrument and is future work. The BigVGAN path currently re-vocodes (running two vocoders); a cleaner design predicts the mel directly, which we leave to v3 along with multi-speaker conditioning (the data has 10 speakers; the single-speaker backbone lacks speaker-conditioning layers). Finally, the base `mms-tts-ara` is released under a non-commercial license, which the derived weights inherit.

13 Conclusion

Shami-TTS turns a promising but robotic Levantine/English code-switching TTS baseline into a production-grade, real-time system on a single consumer GPU. Four targeted changes — a dialectal front-end with automatic diacritization and de-desinentialization, a deterministic duration head, a 24 kHz decoder with an accumulated-adversarial texture recipe, and an anti-aliased BigVGAN vocoder — cut Levantine CER by 86% and WER by 65% while reaching reference-level texture and $\geq 11\times$ real-time speed. We release code, both decoder checkpoints, and audio. The remaining frontier — multi-speaker breadth and a single-pass mel→BigVGAN model — is well scoped for v3.

Reproducibility

Code, configs, checkpoints (HiFi-GAN and BigVGAN pipeline), the 40-clip eval set, and all audio are released. The pipeline is: `prepare_lahgtna.py` (extract + phonemize) → `finetune_gan.py --deterministic-duration --sample-rate 24000 --c-fm 4 --c-stft 3` → `eval_checkpoint.py --auto-ls --asr` → `bigvgan_vocoder.synthesize`. All experiments use a single RTX 3060 (12 GB).

A Levantine G2P: dialect rules

Table 2 lists the headline Levantine realizations encoded in the rule-based G2P. These are the features most audible to a native listener and the ones an MSA G2P gets wrong. The rule engine additionally applies emphatic vowel backing (a,aː→ɑ,ɑː near ق ظ ط ض ص), CCC epenthesis, definite-article sun/moon assimilation, and a high-frequency function-word lexicon.

Table 2: Selected Levantine G2P rules (grapheme / context → IPA), with a spoken example. MSA realizations are shown for contrast.

Grapheme/context	Levantine	(MSA)	Example
ق *qaf*	ʔ	q	قلب→ʔalb “heart”
ج *jim*	ʒ	d͡ʒ	جبنة→ʒibne “cheese”
ث *tha*	t	θ	(urban stop merger)
ذ *dhal*	d	ð	(urban stop merger)
ظ *dha*	zˤ	ðˤ	emphatic
ة final *ta-marb.*	e	a	imala; a after gutturals
ay, aw	eː,oː	aj,aw	بيت→beːt, يوم→joːm
case ending (iʿrab)	∅	-u,-i,-a	de-desinentialization
شو	ʃuː	—	lexicon
بدي	biddi	—	lexicon “I want”
هلق	hallaʔ	—	lexicon “now”

B Front-end worked examples

Table 3 shows the front-end output for representative inputs, including a glued code-switch and a numeral. The language-ID stream (A=Arabic, E=English, ·=neutral/boundary) is emitted in lockstep with the phonemes.

Table 3: Text → IPA (+ language-ID sketch) from the deterministic front-end.

Input	IPA	lang stream
إحجز بَدّي flight بيروت من to London بُكرا	ˈbiddi ʔiħʒiz flˈaɪt min beːruːt tə lˈʌndən bukra	A A E A A E E A
ليش trainingال مِشْ شغّال on the serverؤ	leːʃ ʔil tɹˈeɪnɪŋ miʃ ʃaɣɣaːl ɔnðə sˈɜːvɚ	A A E A A E E
عندي meeting الساعة 3:30	ʕindi mˈiːtɪŋ ʔissaːʕa θalaːθe wnusˁsˁ	A E A A(num)

C Hyperparameters

Table 4: Training configuration (single RTX 3060, 12 GB, bf16).

optimizer	AdamW (0.8, 0.99)	learning rate	2×10^{-4}
batch size	12–16	`seg_size`	8192–12288
c_{mel}	45	c_{kl}	1
c_{dur}	1–2	c_{fm}	4
c_{stft}	3	MR-STFT ffts	512/1024/2048
mel bins	80 (loss)	sample rate	24,000 Hz
hop / n_{fft}	256 / 1024	grad clip	10
steps/stage	8k–10k	save every	1k (atomic)
VITS params	∼36 M	disc. / BigVGAN	47 M / 112 M

D Data composition and per-stage checkpoints

The corpus (`lahgtna-levantine-tts`, CC-BY-4.0) is 50,000 clips / 66.8 h / 24 kHz, 10 speakers (5M/5F), ∼88% pure Levantine and ∼12% synthetic code-switch, partially diacritized. Single-speaker experiments use speaker *Badr* (5,000

clips, 6.2 h after $[0.8, 12]$ s filtering; 40-clip balanced held-out set). Each contribution is a warm-started fine-tune of the previous best; checkpoints (generator + discriminator) are released for baseline, deterministic-duration (16 kHz), 24 kHz, texture, and the BigVGAN pipeline.

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